Date

Subject :

Neural Network

What is? Each neuron is just a container holds number between 0 → 1 called its "activation"

architecture? Data moves from input layer through hidden layers to output //

what is neuron? → Single computational neuron executes logistic regression

→ Sigmoid function → for binary

→ squishes real continous number into range [0 → 1]

Forward Pass: The initial process where input data flows through network using random, unoptimized weights to make initial guess

Backword err propagation: The calculated error passed backwards → update weights → better decision

activation functions: → introduce non-linearity

→ sigmoid: [0 → 1] range output, binary classification

→ Tanh: [-1 → 1] // // , better than sigmoid center around zero

→ ReLU: x output if x > 0 else output 0

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Loss Function: Tracks error of individual Training sample

Cost " : " Cumulative avg err over entire batch

implementation

Vectorization: → use numpy To perform Parallel operation instead of slow For loops

Gradient Descent: (Code structure)

→ Define "epoch" loop

→ compute predicted output

→ Calc loss

→ compute Gradients

→ update weights and bias

Churn Prediction using ANN

→ Preprocessing

→ model arch → Keras.Sequential([keras.layers.Dense()])

→ compile model.compile(optimizer 'adam', loss ' ', metrics[])

→ Train model.fit(x, y, epochs)

→ we use classification report for evaluation